Definition: Let S be a basis of V.

1). If S is a finite set, then the number of vectors in S is called the dimension of V, and V is called a finite-dimensional vector space.

2). If S contains infinitely many vectors, then V is called an infinite-dimensional vector space.

Corollary. Let V be a finite-dimensional vector space with $\dim V = n$.

(1). Any linearly independent subset of V which contains exactly n vectors is a basis of V.

(2) Any subset of V which spans V is a basis of V.

Proof (1: Let $S = \{\vec{v}_1, \cdots, \vec{v}_n\} \subseteq V$ such that S is linearly independent.

Let $\mathcal{B}$ be a basis of V. Then $|\mathcal{B}| = n$.

Use the notation in the replacement theorem, take $G = \mathcal{B}$, and $L = S$.

Then there exists a set H with $|H| = |G| - |L| = n - n = 0$. such that

$\operatorname{Span}(L \cup H) = V$. Here, as $|H| = 0$, $H = \phi$. and $L = S$

$\Rightarrow \operatorname{Span}(L \cup H) = \operatorname{Span} S = V$.

$\Rightarrow$ S is a spanning set of V

$\Rightarrow$ S is a basis of V. // end of Jan 13

(2). Suppose $S = \{\vec{v}_1, \cdots, \vec{v}_n\}$ and $\operatorname{Span} S = V$. Then suppose towards a contradiction that S is linearly dependent. Then there exists $S' \subsetneq S$ such that $\operatorname{Span} S' = V$. Use the notation in the replacement theorem: take $G = S'$ then $|G| < n$, and take L to be any basis of V. Then $|L| = n$.

By replacement theorem: $|G| \geq |L|$ then: $n > |G| \geq |L| = n$, Contradiction.

Remarks: (1) For any linearly independent set L with $|L| < \dim V = \#$ of vectors in any basis.

L cannot span V. $\Rightarrow$ A basis is a minimal spanning set.

(2) For any linear independent set $L \subseteq V$, $|L| \leq \dim V$. $\Rightarrow$ A basis a maximal linearly independent set.

Proof: (1). Suppose towards a contradiction that L is linearly independent set such that $|L| < \dim V$ and $\operatorname{Span} L = V$.

Then take any basis B of V. then $\dim V = |B|$.

By replacement theorem, $\dim V = |B| \geq |L|$. contradiction

(2). Take any basis B of V. then by replacement theorem $|B| = \dim V \geq |L|$. ∎

Proposition. Let V be a finite-dimensional vector space with $\dim V = n$. Let $W \subseteq V$ be a subspace with $\dim W = n$. Then $W = V$.

Proof: Let $B = \{\vec{w}_1, \vec{w}_2, \cdots, \vec{w}_n\}$ be a basis of W.

Suppose towards a contradiction that $W \subsetneq V$. Then $\exists \vec{z} \in V - W$.

Then $\{\vec{w}_1, \cdots, \vec{w}_n, \vec{z}\}$ must be linearly independent. In fact, if $\{\vec{w}_1, \cdots, \vec{w}_n, \vec{z}\}$ were linearly dependent, then there exists $c_1, \cdots, c_n, d$ not all zeros such that $c_1\vec{w}_1 + \cdots + c_n\vec{w}_n + d\vec{z} = \vec{0}$.

① If $d \neq 0$, then. $\vec{z} = -\frac{1}{d}(c_1\vec{w}_1 + \cdots + c_n\vec{w}_n) \in \operatorname{Span}\{\vec{w}_1, \cdots, \vec{w}_n\} = W$, contradiction

or ② If $d = 0$, then $\exists\ c_1, \cdots, c_n$ not all zeros: $c_1\vec{w}_1 + \cdots + c_n\vec{w}_n = \vec{0}$

$\Rightarrow \{\vec{w}_1, \cdots, \vec{w}_n\}$ linearly dependent. Contradiction.

$\Rightarrow$ V contains a linearly independent set $\{\vec{w}_1, \cdots, \vec{w}_n, \vec{z}\}$ with cardinality $\geq \dim$

Contradiction.

Remark: this is not true if V is an infinite dimensional vector space.

For example: Let $V = \mathbb{R}^\infty$ and $W = \{(0, a_1, a_2, \cdots) : \forall a_i \in \mathbb{R}\} \subsetneq V$.

Definition: Let V be a finite-dimensional vector space. Let $B = \{\vec{x}_1, \cdots, \vec{x}_n\} \subseteq V$ be an ordered basis of V.

Recall that $\forall\, \vec{x} \in V$, there exists a unique n-tuple $a_1, \cdots, a_n$ such that

$$\vec{x} = a_1\vec{x}_1 + a_2\vec{x}_2 + \cdots + a_n\vec{x}_n.$$

We define: $[\vec{x}]_B = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{R}^n$ (or $\mathbb{C}^n$) as the coordinates of $\vec{x}$ under the ordered basis B.

Remark: Using the above notation: $\varphi_B : V \to \mathbb{R}^n$, $\vec{x} \mapsto [\vec{x}]_B$ is a one-to-one and onto function.

Moreover, one can also prove that φ_B is a linear transformation

thus, φ_B is a vector space isomorphism, and it gives the corresponding "equivalent" vector space structures between V and $\mathbb{R}^n$.

Examples: 1). $V = \mathbb{R}^n$ (over $\mathbb{R}$) or $V = \mathbb{C}^n$ (over $\mathbb{C}$).

standard basis: $\{\vec{e}_1, \vec{e}_2, \cdots, \vec{e}_n\}$ where $\vec{e}_i = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$ the ith entry is 1, all other entries are 0.

$\Rightarrow$ $\dim V = n$

2). $V = \{(a_1, a_2, \cdots);\ a_i \in \mathbb{R} \text{ (or } \mathbb{C})\} = \mathbb{R}^\infty$ (or $\mathbb{C}^\infty$).

then a basis for V can be given by:

the i-th component

$$\{\vec{d}_i = (0, 0, \cdots, 0, 1, 0, \cdots);\quad i = 1, 2, \cdots\}$$

$\Rightarrow \dim V = \infty$.

3) $V = \mathbb{C}$ over $\mathbb{R}$

standard basis: $\{1,\ i\}$ $\Rightarrow \dim V = 2$

In general $V = \mathbb{C}^n$ over $\mathbb{R}$

standard basis: $\{\vec{e}_1, \cdots, \vec{e}_n, \vec{d}_1, \cdots \vec{d}_n\}$,

where $\vec{e}_k = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$, $\vec{d}_j = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ i \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ the j-th entry is $i = \sqrt{-1}$.

Remark: Unless otherwise pointed out, vectors spaces with entries or coefficients of vectors in $\mathbb{R}$ are over $\mathbb{R}$. Vector spaces with entries or coefficients of vectors in $\mathbb{C}$ are over $\mathbb{C}$.

4). $V = \mathbb{R}^{m\times n}$ or $\mathbb{C}^{m\times n}$

standard basis: $\{E^{(i,j)}:\ 1 \le i \le m,\ 1 \le j \le n\}$

where $E^{(i,j)} \in \mathbb{R}^{m\times n}$ (or $\mathbb{C}^{m\times n}$) with the (i,j)-th entry equal to 1, and all other entries are zeros.

$\Rightarrow \dim V = m\cdot n$

5). $V = \mathbb{P}^n(\mathbb{R})$ or $\mathbb{P}^n(\mathbb{C})$.

Standard basis: $\{1, x, x^2, \cdots, x^n\}$

$\dim V = n+1$

6) $V = \mathbb{P}(\mathbb{R})$ or $\mathbb{P}(\mathbb{C})$

standard basis: $\{1, x, x^2, \cdots \}$

$\Rightarrow$ V is an infinite-dimensional vector space.

7). Let $S = \{A \in \mathbb{R}^{n\times n} : A^T = A\}$, i.e. S is the set of symmetric matrices in $\mathbb{R}^{n\times n}$. Find a basis for S and the dimension of S.

For each $1 \le k < l \le n$:

$$S^{(k,l)} = (s_{ij})_{n\times n} \quad \text{with} \quad s_{ij} = \begin{cases} 1, & \text{if } (i,j) = (k,l) \text{ or } (i,j) = (l,k) \\ 0, & \text{otherwise} \end{cases}$$

For each $1 \le d \le n$:

$$E^{(d,d)} = (e_{ij})_{n\times n} \quad \text{with} \quad e_{ij} = \begin{cases} 1, & \text{if } i=j=d \\ 0, & \text{otherwise.} \end{cases}$$

Let $B = \{ S^{(k,l)} : 1 \le k < l \le n \} \cup \{ E^{(d,d)} : 1 \le d \le n \}$

Exercise: show that B is a basis for S. What is the dimension of S?

8). $W = \{ A \in \mathbb{R}^{n\times n} : \mathrm{Tr}(A) = 0 \}$.

① Show that W is a subspace of $\mathbb{R}^{n\times n}$.

② Find a basis of W.

Solution: ① $\forall A, B \in W$, and $\forall c \in \mathbb{R}$

$\Rightarrow \mathrm{Tr}(A) = \mathrm{Tr}(B) = 0$.

Denote $A = (a_{ij})$, $B = (b_{ij})$

Linear Transformations.

Definition: Let V and W be vector spaces over $\mathbb{R}$ (or over $\mathbb{C}$). A linear transformation $T: V \to W$ is a function satisfying the following conditions:

1): $\forall \vec{x}, \vec{y} \in V$: $\quad T(\vec{x}+\vec{y}) = T(\vec{x}) + T(\vec{y})$

2) $\forall \vec{x} \in V$ and $\forall c \in \mathbb{R}$ (or $\mathbb{C}$): $T(c\vec{x}) = cT(\vec{x})$.

Remarks: 1) and 2) together is equivalent to the following condition:

$\forall \vec{x}, \vec{y} \in V$ and $\forall c \in \mathbb{R}$ (or $\mathbb{C}$): $T(c\vec{x}+\vec{y}) = cT(\vec{x}) + T(\vec{y})$.

Examples: 1) The identity linear transformation on V:

$$\begin{aligned} Id: \ & V \to V \\ & \vec{v} \mapsto \vec{v} \end{aligned}$$

2) The zero linear transformation from V to W:

$$\begin{aligned} T: \ & V \to W \\ & \vec{v} \mapsto \vec{0}. \end{aligned}$$

3). Let $A \in \mathbb{R}^{m \times n}$. The left-multiplication linear transformation:

$$\begin{aligned} L_A: \ & \mathbb{R}^n \to \mathbb{R}^m \\ & \vec{x} \mapsto A\vec{x} \end{aligned}$$

4). Taking transpose of a matrix: $\quad T: \mathbb{R}^{m \times n} \mapsto \mathbb{R}^{n \times m}$

$$A \mapsto A^T$$

Exercise: verify that T is a linear transformation.

Proposition: If $T: V \to W$ is a linear transformation, then $T(\vec{0}_v) = \vec{0}_w$

<u>Proof</u>: Denote $\vec{w} = T(\vec{0}_v) \in W$.

$$\vec{w} = T(\vec{0}_v + \vec{0}_v) = T(\vec{0}_v) + T(\vec{0}_v) = 2\vec{w}$$

$$\Rightarrow \vec{w} = 2\vec{w} \Rightarrow 2\vec{w} - \vec{w} = \vec{0} \Rightarrow \vec{w} = \vec{0} \in W.$$

Example: 1) Translation. Let $\vec{a} \in \mathbb{R}^n$ be a non-zero vector. Define $T_{\vec{a}}: \mathbb{R}^n \to \mathbb{R}^n$

by $T_{\vec{a}}(\vec{x}) = \vec{x} + \vec{a}$. Then $T_{\vec{a}}(\vec{0}) = \vec{a} \neq \vec{0}$

$\Rightarrow T_{\vec{a}}$ is NOT a linear transformation.

2) Let $\vec{a} \in \mathbb{R}^m$ be a non-zero vector. Define $T: \mathbb{R}^n \to \mathbb{R}^m$ by

$T(\vec{x}) = \vec{a}$. Then T is not a linear transformation.

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